

Use Cases

for

Smart Class and Halls Vacancy Sharing System

Version 1.0 approved

Prepared by

Ritesh Reddy

Vamsi

LJ Vishnu

13 February 2026

Revision History

Name	Date	Reason For Changes	Version
Ritesh Reddy	13-02-2026	Initial Draft	1.0
Vamsi	13-02-2026	Initial Draft	1.0
LJ Vishnu	13-02-2026	Initial Draft	1.0

The various user classes identified the following use cases and primary actors for the Campus Management System:

Primary Actor	Use Cases
Faculty	1. Book Classroom 2. Request Extra Class 3. Submit Event Proposal 4. View Event Calendar
Class Representative (CR)	5. Book Classroom 6. Submit Event Proposal 7. View Event Calendar
Classroom Admin	8. Approve Classroom Booking 9. Reject Classroom Booking 10. Manage Classroom Schedule
Seminar Hall Admin	11. Approve Seminar Hall Booking 12. Reject Seminar Hall Booking 13. Manage Seminar Hall Schedule
Event Organizer	14. Create Event 15. Modify Event Details 16. Cancel Event
System Admin	17. Add Classroom Details 18. Add Seminar Hall Details 19. Manage User Accounts 20. Generate System Usage Reports

Use Case 1: Book Classroom

Covers Original Use Cases:

- (1) Book Classroom
- (2) Request Extra Class
- (5) Book Classroom (CR)
- (4) View Event Calendar
- (7) View Event Calendar (CR)

Use Case ID	1
Use Case Name	Book Classroom
Created By	Ritesh Reddy
Last Updated By	Ritesh Reddy
Date Created	13 February 2026
Date Last Updated	13 February 2026
Actors	Faculty, Class Representative (CR)
Description	A Faculty member or Class Representative accesses the Campus Management System to request booking of a classroom for conducting regular classes, extra classes, or academic events on a specified date and time slot.
Preconditions	1. User must be logged into the system. 2. Classroom details must exist in the database. 3. User must have valid booking permission.
Postconditions	1. Booking request is stored with status “Pending”. 2. Notification is sent to the respective Classroom Admin. 3. Time slot remains reserved until approval or rejection.
Normal Flow	1. User selects “Book Classroom” option. 2. System displays available classrooms. 3. User selects classroom, date and time slot. 4. System checks classroom availability. 5. System displays availability status. 6. User enters purpose (Regular Class / Extra Class / Event). 7. User submits booking request. 8. System stores request with status “Pending”. 9. System forwards request to Classroom Admin. 10. System displays confirmation message.

Alternative Flows	1.1 Modify Booking (after step 3): - User changes date or time. - System rechecks availability. - Return to step 5. 1.2 Cancel Booking (before step 7): - User cancels request. - System terminates use case.
Exceptions	1.0.E.1 Classroom Not Available (at step 4): - System informs classroom is already booked. - User selects different slot. 1.0.E.2 Invalid Time Selection: - System displays error message. - User re-enters valid time. 1.0.E.3 Unauthorized Access: - System denies access. - Use case terminates.
Includes	Check Venue Availability Send Notification
Priority	High
Frequency of Use	Frequently during academic sessions
Business Rules	- No classroom shall be double-booked for the same time slot. - Booking is confirmed only after admin approval.
Special Requirements	- System shall perform real-time availability checking. - Approved bookings shall automatically update the Event Calendar.
Assumptions	- All classrooms are pre-registered in the system. - Each classroom has an assigned admin.
Notes and Issues	Recurring weekly booking may be supported in future versions.

Use Case 2: Approve / Reject Classroom Booking

Covers Original Use Cases:

- (8) Approve Classroom Booking
- (9) Reject Classroom Booking
- (10) Manage Classroom Schedule

Use Case ID	2
Use Case Name	Approve / Reject Classroom Booking
Created By	Vamsi
Last Updated By	Vamsi
Date Created	13 February 2026
Date Last Updated	13 February 2026
Actors	Classroom Admin
Description	The Classroom Admin reviews pending classroom booking requests submitted by Faculty or CR and decides to approve or reject the request based on availability and scheduling rules.
Preconditions	1. Admin must be logged into the system. 2. Booking request must exist with status “Pending”.
Postconditions	1. Booking status is updated to “Approved” or “Rejected”. 2. User is notified of the decision. 3. If approved, the classroom schedule is updated.
Normal Flow	1. Admin logs into the system. 2. Admin selects “View Pending Requests”. 3. System displays list of pending classroom bookings. 4. Admin selects a booking request. 5. System displays booking details. 6. Admin reviews date, time, and purpose. 7. Admin selects “Approve” option. 8. System updates booking status to “Approved”. 9. System updates classroom schedule. 10. System sends notification to the requester.
Alternative Flows	2.1 Reject Booking Request (at step 7): a) Admin selects “Reject” option. b) Admin provides rejection reason. c) System updates status to “Rejected”. d) System notifies requester.

Exceptions	2.0.E.1 Booking No Longer Available (at step 6): a) System detects scheduling conflict. b) Admin cannot approve request. 2.0.E.2 Unauthorized Access: a) System denies access. b) Use case terminates.
Includes	Update Classroom Schedule Send Notification
Priority	High
Frequency of Use	Daily during academic sessions
Business Rules	1. Only assigned Classroom Admin can approve or reject requests. 2. Approval must not violate existing schedule conflicts.
Special Requirements	1. System shall maintain audit log of approval decisions. 2. Notifications shall be sent immediately after decision.
Assumptions	1. Admin has authority over assigned classrooms.
Notes and Issues	1. Future enhancement may include auto-approval for recurring bookings.

Use Case 3: Book Seminar Hall

Covers Original Use Cases:

- Seminar Hall Booking Request
- (14) Create Event

Use Case ID	3
Use Case Name	Book Seminar Hall
Created By	LJ Vishnu
Last Updated By	LJ Vishnu
Date Created	13 February 2026
Date Last Updated	13 February 2026
Actors	Faculty, Event Organizer
Description	A Faculty member or Event Organizer requests booking of a seminar hall or auditorium for conducting seminars, workshops, guest lectures, or other academic events on a specified date and time slot.
Preconditions	1. User must be logged into the system. 2. Seminar hall details must exist in the system database. 3. User must have valid booking permissions.
Postconditions	1. Booking request is stored with status “Pending”. 2. Request is forwarded to the respective Seminar Hall Admin. 3. User receives confirmation of request submission.
Normal Flow	1. User selects “Book Seminar Hall” option. 2. System displays list of available seminar halls. 3. User selects desired hall. 4. User selects date and time slot. 5. System checks hall availability. 6. System displays availability status. 7. User enters event details (Event Name, Purpose, Expected Participants). 8. User submits booking request. 9. System stores request with status “Pending”. 10. System forwards request to Seminar Hall Admin. 11. System displays confirmation message.
Alternative Flows	3.1 Modify Booking Details (after step 4): a) User changes date or time slot. b) System rechecks availability. c) Process returns to step 6. 3.2 Cancel Booking Request (before step 8): a) User cancels request. b) System terminates use case.

Exceptions	3.0.E.1 Seminar Hall Not Available (at step 5): a) System informs user that hall is already booked. b) User selects another time slot or hall. 3.0.E.2 Invalid Event Details: a) System displays validation error. b) User corrects event information. 3.0.E.3 Unauthorized Access: a) System denies access. b) Use case terminates.
Includes	Check Venue Availability Send Notification
Priority	High
Frequency of Use	Frequently during academic and event periods
Business Rules	1. Seminar hall shall not be double-booked. 2. Booking must be approved by Seminar Hall Admin before confirmation.
Special Requirements	1. System shall allow entry of detailed event information. 2. Approved bookings shall automatically update the Event Calendar.
Assumptions	1. All seminar halls are registered in the system. 2. Each hall has an assigned admin for approval.
Notes and Issues	1. Future enhancement may include resource request (Projector, Sound System).

Use Case 4: Approve / Reject Seminar Hall Booking

Covers Original Use Cases:

- (11) Approve Seminar Hall Booking
- (12) Reject Seminar Hall Booking
- (13) Manage Seminar Hall Schedule

Use Case ID	4
Use Case Name	Approve / Reject Seminar Hall Booking
Created By	LJ Vishnu
Last Updated By	LJ Vishnu
Date Created	13 February 2026
Date Last Updated	13 February 2026
Actors	Seminar Hall Admin
Description	The Seminar Hall Admin reviews pending seminar hall booking requests submitted by Faculty or Event Organizers and decides to approve or reject the request based on availability, scheduling constraints, and institutional policies.
Preconditions	1. Admin must be logged into the system. 2. Seminar hall booking request must exist with status “Pending” .
Postconditions	1. Booking status is updated to “Approved” or “Rejected”. 2. If approved, the seminar hall schedule is updated. 3. Requester is notified about the decision.
Normal Flow	1. Admin logs into the system. 2. Admin selects “View Pending Seminar Hall Requests”. 3. System displays list of pending requests. 4. Admin selects a request to review. 5. System displays event details and requested time slot. 6. Admin verifies availability and event details. 7. Admin selects “Approve”. 8. System updates booking status to “Approved”. 9. System updates seminar hall schedule. 10. System sends notification to requester.
Alternative Flows	4.1 Reject Booking Request (at step 7): a) Admin selects “Reject”. b) Admin enters rejection reason. c) System updates status to “Rejected”. d) System notifies requester.

Exceptions	4.0.E.1 Scheduling Conflict Detected (at step 6): a) System identifies conflict in schedule. b) Admin cannot approve request. 4.0.E.2 Unauthorized Access: a) System denies access to approval module. b) Use case terminates.
Includes	Update Seminar Hall Schedule Send Notification
Priority	High
Frequency of Use	Frequently during event scheduling periods
Business Rules	1. Only assigned Seminar Hall Admin can approve or reject bookings. 2. No double booking of seminar halls is allowed.
Special Requirements	1. System shall maintain an audit log of approval decisions. 2. Notifications shall be sent immediately after decision.
Assumptions	1. Seminar hall admins are assigned to specific halls.
Notes and Issues	1. Future enhancement may include automatic approval for low-priority events.

Use Case 5: Submit Event Proposal

Covers Original Use Cases:

- (3) Submit Event Proposal
- (6) Submit Event Proposal (CR)

Use Case ID	5
Use Case Name	Submit Event Proposal
Created By	Ritesh Reddy
Last Updated By	Ritesh Reddy
Date Created	13 February 2026
Date Last Updated	13 February 2026
Actors	Faculty, Class Representative (CR), Event Organizer
Description	A Faculty member, CR, or Event Organizer submits an event proposal through the Campus Management System by providing event details such as event name, description, expected participants, preferred venue, and date/time for approval by the respective venue admin.
Preconditions	1. User must be logged into the system. 2. User must have permission to organize events.
Postconditions	1. Event proposal is stored in the system with status “Pending Approval”. 2. Proposal is forwarded to the respective venue admin. 3. User receives confirmation of proposal submission.
Normal Flow	1. User selects “Submit Event Proposal” option. 2. System displays event proposal form. 3. User enters event details (Event Name, Description, Date, Time). 4. User selects preferred venue (Classroom / Seminar Hall). 5. System checks preliminary availability of selected venue. 6. User confirms submission of proposal. 7. System stores proposal with status “Pending Approval”. 8. System forwards proposal to respective venue admin. 9. System displays confirmation message.
Alternative Flows	5.1 Modify Proposal Before Submission (before step 6): a) User edits event details. b) System updates information. 5.2 Cancel Proposal Submission: a) User cancels proposal. b) System terminates use case.

Exceptions	5.0.E.1 Venue Not Available (at step 5): a) System informs user of conflict. b) User selects different venue or time. 5.0.E.2 Incomplete Event Details: a) System displays validation error. b) User provides missing information. 5.0.E.3 Unauthorized Access: a) System denies access. b) Use case terminates.
Includes	Check Venue Availability Send Notification
Priority	High
Frequency of Use	Moderate to high during academic events
Business Rules	1. Event must be approved by respective venue admin before confirmation. 2. Events must not conflict with existing bookings.
Special Requirements	1. System shall allow entry of detailed event description. 2. All event proposals shall be recorded for auditing purposes.
Assumptions	1. Users organizing events have appropriate institutional permissions.
Notes and Issues	1. Future enhancement may include resource request (Projector, Sound System).

Use Case 6: Manage Event (Modify / Cancel Event)

Covers Original Use Cases:

- (15) Modify Event Details
- (16) Cancel Event

Use Case ID	6
Use Case Name	Manage Event (Modify / Cancel Event)
Created By	Vamsi
Last Updated By	Vamsi
Date Created	13 February 2026
Date Last Updated	13 February 2026
Actors	Event Organizer, Faculty, Class Representative (CR)
Description	An Event Organizer, Faculty member, or CR manages an existing event by modifying event details such as date, time, venue, or description, or cancels the event before its scheduled date.
Preconditions	1. User must be logged into the system. 2. Event must already exist in the system. 3. User must have permission to manage the event.
Postconditions	1. Event details are updated in the system if modified. 2. Event status is updated to “Cancelled” if cancelled. 3. Relevant users are notified about the change.
Normal Flow (Modify Event)	1. User selects “Manage My Events”. 2. System displays list of events created by the user. 3. User selects an event to modify. 4. System displays existing event details. 5. User updates required fields (date, time, venue, description). 6. System checks availability if venue or time is changed. 7. User submits updated details. 8. System saves changes. 9. System notifies concerned users about modification.
Alternative Flow (Cancel Event)	6.1 Cancel Event (after step 3): a) User selects “Cancel Event”. b) System asks for confirmation. c) User confirms cancellation. d) System updates event status to “Cancelled”. e) System notifies participants and venue admin.

Exceptions	6.0.E.1 Venue Conflict After Modification (at step 6): a) System detects scheduling conflict. b) User must select different time or venue. 6.0.E.2 Unauthorized Access: a) System denies access to event management. b) Use case terminates.
Includes	Check Venue Availability Update Event Details Send Notification
Priority	Medium to High
Frequency of Use	Moderate during academic and event periods
Business Rules	1. Only event creator or authorized personnel can modify or cancel event. 2. Cancelled events must release reserved venue automatically.
Special Requirements	1. System shall log all modifications for audit purposes. 2. Cancellation notifications must be sent immediately.
Assumptions	1. Events are associated with a specific venue booking.
Notes and Issues	1. Future enhancement may include rescheduling option without full cancellation.

Use Case 7: Manage System (Venue & User Management)

Covers Original Use Cases:

- (17) Add Classroom Details
- (18) Add Seminar Hall Details
- (19) Manage User Accounts
- (20) Generate System Usage Reports

Use Case ID	7
Use Case Name	Manage System (Venue & User Management)
Created By	LJ Vishnu
Last Updated By	LJ Vishnu
Date Created	13 February 2026
Date Last Updated	13 February 2026
Actors	System Admin
Description	The System Admin manages core system configurations including adding or updating classroom and seminar hall details, creating and managing user accounts, assigning roles, and maintaining system records.
Preconditions	1. Admin must be logged into the system. 2. Admin must have system-level privileges.
Postconditions	1. Venue details are added or updated in the system. 2. User accounts are created, updated, or removed. 3. Role assignments are updated successfully.
Normal Flow (Manage Venues)	1. Admin selects “Manage Venues”. 2. System displays list of existing classrooms and seminar halls. 3. Admin selects “Add” or “Edit” venue. 4. Admin enters or updates venue details (Name, Capacity, Location). 5. Admin saves changes. 6. System updates venue information in database.
Alternative Flow (Manage Users)	7.1 Manage User Accounts: a) Admin selects “Manage Users”. b) System displays list of registered users. c) Admin adds new user or updates existing user details. d) Admin assigns role (Faculty / CR / Admin). e) System saves user information.

Exceptions	7.0.E.1 Duplicate Venue Entry: a) System detects duplicate venue name. b) Admin must enter unique details. 7.0.E.2 Invalid User Information: a) System displays validation error. b) Admin corrects information. 7.0.E.3 Unauthorized Access: a) System denies access. b) Use case terminates.
Includes	Validate Input Update Database Records
Priority	High
Frequency of Use	Occasional (Administrative operations)
Business Rules	1. Only System Admin can modify venue and user records. 2. Each user must have a defined role in the system.
Special Requirements	1. System shall maintain audit logs for all administrative actions. 2. Data consistency must be ensured during updates.
Assumptions	1. System Admin has full system control privileges.
Notes and Issues	1. Future enhancement may include automated role-based access control policies.